

ISM 6251 and 4543: Machine Learning (Fall 2025)

Assignment 3

- Please provide succinct answers to the questions below.
- You should submit a .ipynb file and a pdf (for any non-coding questions) in Canvas.
- Please include the names of all team-members in your write up and in the name of the file. One submission per group.
- You should include all the Python functions you use.

This assignment is developed based on a case study on predicting earnings manipulation from [Harvard Business Review](#). Please read the paragraph below and answer the following questions.

“MCA Technology Solutions Private Limited was established in 2015 in Bangalore with an objective to integrate analytics and technology with business. MCA Technology Solutions helped its clients in areas such as customer intelligence, forecasting, optimization, risk assessment, web analytics, and text mining and cloud solutions. Risk assessment vertical at MCA technology solutions focused on problems such as fraud detection and credit scoring. Sachin Kumar, Director at MCA Technology Solutions, Bangalore was approached by one of his clients, a commercial bank, to assist them in detecting earnings manipulators among the bank's customers. The bank provided business loans to small and medium enterprises and the value of loan ranged from INR 10 million to 500 million. The bank suspected that its customers may be involved in earnings manipulations to increase their chance of securing a loan. Saurabh Rishi, the chief data scientist at MCA Technologies was assigned the task of developing a use case for predicting earnings manipulations. He was aware of models such as Benford's law and Beneish model used for predicting earnings manipulations; however, he was not sure of its performance, especially in the Indian context. Saurabh decided to develop his own model for predicting earnings manipulations using data downloaded from the Prowess database maintained by the Centre of Monitoring Indian Economy (CMIE). Daniel received information related to earning manipulators from Securities Exchange Board of India (SEBI) and the Lexis Nexis database. Data on more than 1200 companies was collected to develop the model. MCA Technology believed that machine learning algorithms may give better accuracy compared to other traditional models such as Beneish model used for predicting earnings manipulation.”

1. The number of manipulators is usually much less than non-manipulators. What kind of modeling problems can one expect when cases in one class are much lower than the other class in a binary classification problem? How can one handle these problems?

When one class (earnings manipulators) is much smaller than the others (non-manipulators), the dataset is imbalanced, and this leads to several modeling challenges, such as biased model predictions, poor recall and precision for the minority class, and misleading evaluation metrics. Most machine-learning algorithms try to minimize overall error. With heavy imbalance, a model can get very high accuracy by only predicting the majority class (non-manipulators), thus leading to poor detection of the minority class. This can be handled by standard data science and collection methods, such as oversampling the minority class in addition to undersampling the majority class to mitigate the imbalance. Furthermore, at the algorithm level, there is also the option of applying cost-sensitive learning or class weighting to assign higher misclassification costs to manipulators.

2. Use `sample_data.csv` (220 cases including 39 manipulators) and develop a logistic regression model that can be used by MCA Technologies Private Limited for predicting probability of earnings manipulation. Describe each step of building the model and explain the rationale when necessary.
3. Comment on the model you developed by explaining the results; how do you measure the accuracy of the model?
4. What should be the strategy adopted by MCA Technology Solutions to deploy the logistic regression

model developed?

Even though logistic regression is conceptually simple and transparent, its deployment in a live financial decision environment involves careful attention to data assumptions, numerical stability, interpretability, and performance consistency.

Logistic regression coefficients are valid only if the data distribution at deployment matches the training data. If financial ratios (e.g., DSRI, AQI, SGI) in new loan applicants differ significantly from the training sample, predictions will be biased. Therefore, MCA should consider implementing data validation checks before scoring (e.g., min/max and z-score thresholds). If drift exceeds acceptable limits, retrain or recalibrate the model.

Additionally, multicollinearity between predictors should be assessed. Financial ratios like DSRI, AQI, and SGI may be correlated (e.g., fast sales growth might drive receivables). Multicollinearity can inflate coefficient variances and destabilize odds ratio interpretations. Therefore, MCA should examine Variance Inflation Factors (VIF) for each predictor pre-deployment. If multicollinearity is high, consider removing or combining correlated variables.

Finally, the 0.5 cutoff used for binary classification may not align with business needs (e.g., identifying manipulators at low prevalence). An uncalibrated threshold can misclassify high-risk firms as “safe.”. That being said, MCA should use the ROC and Precision-Recall curves to find an optimal threshold (e.g., maximize F1-score or minimize cost). Storing threshold as a configurable parameter allows tuning without retraining.

5. Based on the models developed in questions 4 and 5, suggest a Mscore (Manipulator score) that can be used by regulators to identify potential manipulators.

$$\text{M-Score} = \beta_0 + 2.247(\text{DSRI}) + 1.512(\text{GMI}) + 1.692(\text{AQI}) + 1.407(\text{SGI}) + 0.031(\text{DEPI}) - 0.023(\text{SGAI}) + 0.723(\text{ACCR}) - 0.104(\text{LEVI})$$

$$P(\text{Manipulator}) = \frac{1}{1 + e^{-\text{MScore}}}$$

Putting in ranges,

$$P < 0.3 \rightarrow \text{Non-manipulator}$$

$$0.3 \leq P < 0.6 \rightarrow \text{Possible manipulator}$$

$$P \geq 0.6 \rightarrow \text{Likely manipulator}$$

6. Develop classification and regression tree (CART) model. Describe each step of building the model and explain the rationale when necessary. What insights do you obtain from the CART model?
7. Develop a logistic regression model using the *complete_data.csv* (1200 non-manipulators and 39 manipulators), compare the results with the previous logistic regression model.
8. Develop models using machine learning algorithms such as random forest and boosting (or even more than these two). Using appropriate evaluation approaches to compare the outputs from these methods with logistic regression and classification tree.
9. What will be your final recommendation for predicting earnings manipulators? Explain your reasons.

I would pick Gradient Boosting, because it has the highest accuracy of 92.7%, and a ROC-AUC of 0.9. The best way however would be to use gradient boosting to flag any manipulators, then use logistic regression to calculate any risk scores